TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Autonomous and Remote Operation of Microreactors			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources			
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	Nuclear Plant Operation & Maintenance		
Concept of Proposed Technology	Due to their intended deployment in remote or extreme environments, microreactors inherently necessitate autonomous operational capabilities. This approach is augmented by remote monitoring and emergency override protocols to ensure safety and reliability. By leveraging high-fidelity microreactor simulations, plant-state mapping, and operational trajectory prediction techniques, this project provides the mathematical foundation and licensing evidence required for robust autonomous and remote operation strategies of the target microreactor.			
Rationale for Joint R&D Project	Through a strategic partnership, we will identify a target microreactor and develop its dedicated simulation environment. Using RPI's plant-state mapping and operational trajectory prediction techniques, we will conduct synchronized autonomous and remote operation experiments across two geographic locations (New York and Korea). This international dual-site approach ensures systematic validation of the proposed operational strategies under diverse conditions.			
Key Words	1. Autonomous operation of microreactor 2. Plant state mapping and prediction			

	3. Remote operation experiments

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)
R&D Objectives	- Develop autonomous control algorithms for the required functions of the target microreactor - Design high-level control architecture for autonomous and remote control of microreactor based on the system response domain analysis - Demonstrate safety evidence of the target microreactor in remote-autonomous operation environment			
R&D Target Specifications	- Advanced control algorithms which ensure the target plant stays in the allowable operational area defined by precision and time - Cross-border remote operation demonstration between South Korea and New York - Safety evidence collection from the experiments with more than 50 sampled plant transient cases considering risk priority			
R&D Plan	<u>Background</u> Due to their unique operational requirements, for microreactors, remote and autonomous operation is not optional, it is a foundational licensing issue affecting: - Staffing requirements - Emergency response assumptions - Cybersecurity classification - Human factors engineering - Defense-in-depth philosophy Regulators (US NRC and Korean NSSC) will require: - Empirical evidence - Integrated system-level validation - Demonstrated fail-safe behavior under abnormal conditions This R&D project aims at the development of autonomous and remote operation of micro nuclear reactors. The focus of research will explore the feasibility of the related architecture design and demonstrate the applicability to the target microreactor plant. Detailed plant design and operational events coverage can be addressed in the future. <u>Approach</u> - Since the target microreactor is not physically available at the time of this project, it should be computationally simulated. Modelica simulation will be used. Dymola (Dassault Systems) programing environment will be utilized for the enhanced user interface and the extended library. TRANSFORM library (Oak Ridge National Lab) and in-house developed models for various nuclear reactors and associate energy transfer components will be used. - Remote operation combined with autonomous operation requires detailed system response analysis. In most operational demand cases, the autonomous operational algorithms will control the system states, but in some specific cases, the remote operation action must be incorporated through override mechanism. Systematic system response analysis will provide			

a system state map and dynamic transition probabilities among the states. An in-house developed framework shown in Figure 1 will be able to provide mathematical basis of this complex analysis.

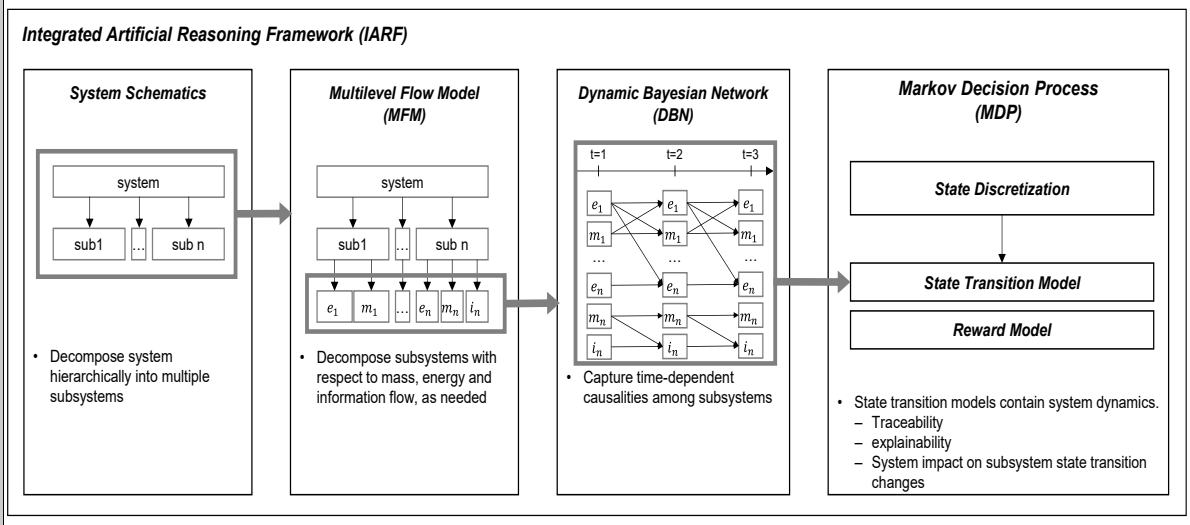

Figure 1. An example integrated reasoning framework [1]: System configuration was converted into the relationship among physical parameters, then Bayesian network for each time step was constructed based on state transition probabilities, which were used for Markov decision making process.

- How the remote operation override can be seamlessly combined with autonomous operation scheme and how the integrated operational framework can be validated are crucial issues in microreactor licensing, thus it requires a systematic analysis on licensing process of advanced reactor. Licensability should be carefully managed.
- Basic graphical user interface for the remote operation will be developed by using engineering tools like LabView. To demonstrate cross-border remote operation, it will be installed on the engineering consoles of remote places like South Korea and New York while the plant virtually locates in another place. Various aspects of this experiment, such as operator action time delays, operator's cognitive load, and cross-bord collaboration efficiency, will be systematically analyzed.
- For the transient scenario development, both external and internal events will be considered as initiating events. Possible scenarios following a specific initiating event can be captured by conventional techniques like event tree analysis.

<u>Tasks</u>

Task 1: Development of microreactor simulation models

- Target microreactor design will be collaboratively identified
- Modelica language will be used for simulation for the microreactor core and energy transfer components.
- External environments will be assumed to be stable.
- In-house component models, open sources for general components, and public or commercial library will be fully used.

Task 2: System response domain dynamic analysis

- Plant response under various initiating events will analytically be modeled based on the physical relations of process properties to identify the process parameters which determine

	the plant state. The response domain will be simplified to reduce computational burden. - Once the parameters are identified, dynamic plant state mapping will be performed based on the simulation data. The transition probabilities among the possible plant states will also be determined based on the simulation results. - The sequence of analysis of IARF in Figure 1 will be used as a basis of this analysis. Task 3: Architecture of autonomous operation with remote override mechanism - This task is to define how the remote operation override can be seamlessly combined with autonomous operation scheme. - The cases where autonomous operational algorithms will control plant systems without remote decision making should be defined by process parameters. - For specific cases when process parameters indicate the remote operation action must be incorporated, there should be an override mechanism. This requires careful analysis for seamless connection with existing autonomous control mechanism. Task 4: Control algorithms for autonomous operation - For the major components of the target plant, the dedicated control algorithm will be developed. - The focus of this development should be to achieve autonomous operation for anticipated operational occurrences rather than the accuracy or performance of individual controllers. Task 5: Integration of microreactor model with autonomous control algorithms - Developed control algorithms in Task4 should be implemented to the computational plant model developed in Task1. - Integration test with selected transient scenario should be performed to validate the developed control algorithms. Task 6: Remote operation user interface and connection to the integrated model - This task is to design an operator interface optimized for remote supervision, not local control rooms. - Unique characteristics of this remote operation including reduced information displays, cognitive workload management, alarm prioritization for remote contexts, and multi-reactor supervision capability should be considered. Task 7: Initiating events and transient scenario - Considering the scope of this R&D project, the target initiating events must be selected. This target event list should include operational occurrence, demand changes, and accidents. While the full scope scenario coverage is not the goal of this study, to demonstrate the unique feature of autonomous operation in conjunction with remote override, both normal and abnormal events must be included in the selected initiating events. - At a specific initiating event, the following operational scenario should be developed. Task 8: Demonstration experiments for remote operation - The developed package, which includes plant simulation, autonomous control, and remote override mechanism, will be examined in this experiment for the transient scenarios developed in Task7. - To demonstrate cross-border remote operation, this package will be installed on the engineering consoles of remote places like South Korea and New York while the plant virtually locates in another place. - Cybersecurity measures could be considered in this task, but the development of novel techniques for cyber secured communication is not the goal of this project.

	Task 9: Experiment results analysis and licensability review - The results of remote operation test campaigns will be collected and analyzed in this task. - Engineering insights for enhanced plant design in terms of both safety and operability will be produced from this analysis. - Production of concrete and reviewable evidence for licensing engagement is the target. <u>Deliverables with Timeline</u> Deliverable 1: Microreactor simulation package (Year 1) Deliverable 2: Conceptual architecture design for autonomous operation with remote override mechanism (Year 1) Deliverable 3: Control algorithms for autonomous operation (Year 2) Deliverable 4: User interface for remote operation (Year 2) Deliverable 5: Integrated package of microreactor simulation, control, and remote operation (Year 3) Deliverable 6: Remote operation experiments analysis report (Year 3) Deliverable 7: Final report (Year 3)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI (PI: Hyun Gook Kang) will be leading efforts for Deliverables 1, 3, and 5. - RPI will develop Modelica simulation model for the target microreactor plant by utilizing modeling libraries, in-house models, and other available resources. - RPI will support the Korean collaborator for the conceptual design of autonomous control with remote override mechanism by using the in-house developed state mapping technique. - Development of detailed control algorithms for major components of target microreactor will be performed. - The developed simulation model will be connected to the control algorithms and remote operational user interface to make a final package. - RPI and the Korean collaborator will work together for the remote operation test campaign. - The experimental results will be analyzed in a systematic framework. Deliverables 6 and 7 will be collaboratively produced.
Expected R&R of Korean SME Partner	Korean companies are expected to lead efforts for Deliverables 2 and 4. - Target micro nuclear reactor information will be prepared. - Will lead the architecture design for autonomous operation with remote override mechanism considering the characteristics of the target microreactor. - User interface for remote operation capability will be developed - RPI and the Korean collaborator will work together for the remote operation test campaign. - Licensability of the developed architecture will be reviewed based on the remote-control experimental campaign and its analysis results Deliverables 6 and 7 will be collaboratively produced.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Microreactors have their own unique operational requirements due to its size and inherent safety features. In most cases they will be deployed to remote or harsh environmental sites. Thus, their remote and autonomous operation is not optional, it is a foundational licensing issue. This work is targeting at producing evidence of autonomous operation with remote override mechanisms. It is expected to serve as concrete and reviewable evidence for licensing engagement. Considering there are many companies worldwide working on microreactor development, this platform and experimental results will serve as a basis of foundation of commercialization, not only as a part of microreactor system package but also as an independent toolkit.

6. References

(1) Junyung Kim, Xinyan Wang, Xingang Zhao, Birdy Phathanapirom, Kyle Warns, Hyun Gook Kang, Michael Golay, “Decision-making based on Markov decision process in integrated artificial reasoning framework - Part 1: Theory,” Nuclear Science and Technology Open Research, 2024.

(2) Kyle Warns, Junyung Kim, Xinyan Wang, Xingang Zhao, Birdy Phathanapirom, Hyun Gook Kang, Michael Golay, “Decision-making based on Markov decision process in integrated artificial reasoning framework - Part 2: Applications,” Nuclear Science and Technology Open Research, Revised in 2025

(3) In Seop Jeon and Hyun Gook Kang, “Development of an optimal mitigation strategy by estimating the conditional core damage probability with time-dependent recovery actions,” Annals of Nuclear Energy, Vol. 142, 2020

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization